0.1 Semi-Implicit (scheme 1)

$$
\begin{aligned}
\frac{1}{\Delta \mathrm{t}}((h\mathbf{u})' - h^{(n)}\mathbf{u}^{(\mathrm{n})}) &= -\nabla \cdot \left(h^{(n)}\mathbf{u}^{(\mathrm{n})}(1-\alpha)\mathbf{u}^{(n)} + h^{i-1}\mathbf{u}^{\mathrm{i}-1}\alpha\mathbf{u}^{i-1})\right) \\
&\quad - h^{(n)}\mathbf{F} \times ((1-\alpha)\mathbf{u}^{(n)}) - h^{i-1}\alpha\mathbf{F} \times \mathbf{u}^{i-1} \\
&\quad - (h^{(n)}(1-\alpha) + h^{i-1}\alpha)g\nabla h_0 \\
\frac{1}{\Delta \mathrm{t}}(h^i - h^{(n)}) &= -\nabla \cdot \left((1-\alpha)h^{(n)}\mathbf{u}^{(\mathrm{n})} + \alpha(h\mathbf{u})'\right) \\
&\quad + \alpha^2(\Delta t)h^{i-1}g\nabla^2(h^i) \\
\frac{1}{\Delta t}(h^i\mathbf{u}^i - (h\mathbf{u})') &= -h^{i-1}\alpha g\nabla(h^i)
\end{aligned}
$$

0.2 Segregated Implicit Advection, Implicit Gravity Waves (scheme 2)

$$
\begin{aligned}
\frac{1}{\Delta \mathrm{t}}(h^{i-1}\mathbf{u}' - h^{(n)}\mathbf{u}^{(\mathrm{n})}) &= -\nabla \cdot \left(h^{(n)}\mathbf{u}^{(\mathrm{n})}(1-\alpha)\mathbf{u}^{(n)} + h^{i-1}\mathbf{u}^{\mathrm{i}-1}\alpha\mathbf{u}')\right) \\
&\quad - h^{(n)}\mathbf{F} \times ((1-\alpha)\mathbf{u}^{(n)}) - h^{i-1}\alpha\mathbf{F} \times \mathbf{u}^{i-1} \\
&\quad - h^{(n)}g\nabla(1-\alpha)(h^{(n)} + h_0) - h^{i-1}g\nabla\alpha(h^{i-1} + h_0) \\
\frac{1}{\Delta \mathrm{t}}(h^{i-1}\mathbf{u}'' - h^{(n)}\mathbf{u}^{(\mathrm{n})}) &= -\nabla \cdot \left(h^{(n)}\mathbf{u}^{(\mathrm{n})}(1-\alpha)\mathbf{u}^{(n)} + h^{i-1}\mathbf{u}^{\mathrm{i}-1}\alpha\mathbf{u}')\right) \\
&\quad - h^{(n)}\mathbf{F} \times ((1-\alpha)\mathbf{u}^{(n)}) - h^{i-1}\alpha\mathbf{F} \times \mathbf{u}^{i-1} \\
&\quad - h^{(n)}g\nabla(1-\alpha)(h^{(n)} + h_0) \\
\frac{1}{\Delta \mathrm{t}}(h^i - h^{(n)}) &= -\nabla \cdot \left(h^{i-1}\mathbf{u}''\right) + \alpha^2(\Delta t)h^{i-1}g\nabla^2(h^i + h_0) \\
\frac{1}{\Delta t}(\mathbf{u}^i - \mathbf{u}'') &= -\alpha g\nabla(h^i + h_0)
\end{aligned}
$$

0.3 Operator Split Implicit Advection, Implicit Gravity Waves (scheme 3)

$$
\begin{aligned}
\frac{1}{\Delta \mathrm{t}}(h^{i-1}\mathbf{u}' - h^{(n)}\mathbf{u}^{(\mathrm{n})}) &= -\nabla \cdot \left(h^{(n)}\mathbf{u}^{(\mathrm{n})}(1-\alpha)\mathbf{u}^{(n)} + h^{i-1}\mathbf{u}^{\mathrm{i}-1}\alpha\mathbf{u}')\right) \\
&\quad - h^{(n)}\mathbf{F} \times ((1-\alpha)\mathbf{u}^{(n)}) - h^{i-1}\alpha\mathbf{F} \times \mathbf{u}^{i-1} \\
\frac{1}{\Delta \mathrm{t}}(h^i - h^{(n)}) &= -\nabla \cdot \left((1-\alpha)h^{(n)}\mathbf{u}^{(\mathrm{n})} + \alpha h^i\mathbf{u}'\right) \\
&\quad + \alpha(\Delta t)h^{(n)}g\nabla^2(1-\alpha)(h^{(n)} + h_0) - \alpha(\Delta t)h^{i-1}g\nabla^2\alpha(h^i + h_0) \\
\frac{h^i}{\Delta t}(\mathbf{u}^i - \mathbf{u}') &= -h^{(n)}(1-\alpha)g\nabla(h^{(n)} + h_0) - h^i\alpha g\nabla(h^i + h_0)
\end{aligned}
$$